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Key Points:

- SHapley Additive exPlanation analysis reveals that temperature is the dominant factor in seasonal $\delta v/v$ across slope types, showing a robust positive correlation
- The correlation exhibits a contrasting pattern: a linear relationship on the rock slope versus a nonlinear trend on the clastic-soil slope
- The discrepancy is more plausibly attributed to the inherent material properties across different types of slopes

Supporting Information:

Supporting Information may be found in the online version of this article.

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Decoupling Seasonal Seismic Velocity Changes on Slow-Moving Slopes in Southwest China Using Tree Ensemble Machine Learning

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Abstract Changes in seismic properties can help assess slope damage, which is influenced by external factors such as temperature, relative humidity, rainfall, air pressure, and local seismicity. However, the contributions of these factors, and how the dominant factor modulates transient and seasonal seismic velocity changes ($\delta v/v$), remain unclear across different slope types. We conduct 2 years of in situ monitoring of daily $\delta v/v$ on two representative slopes in southwest China: the Pubugou (PBG) rock slope and the Jinpingzi (JPZ) clastic-soil slope. We apply an eXtreme Gradient Boosting (XGBoost) regressor and SHapley Additive exPlanation (SHAP) to quantify each factor's marginal contribution. The XGBoost model outperforms other common models in predicting $\delta v/v$. SHAP analysis highlights temperature as the most influential factor while revealing a contrasting pattern, a linear correlation on the PBG slope versus a nonlinear relationship on the JPZ slope. This discrepancy is primarily attributed to the inherent material properties across slope types.

Plain Language Summary Real-time monitoring of damage evolution on slow-moving slopes has been achieved through seismic velocity changes ($\delta v/v$). The changes are influenced by multiple external factors, including air temperature, relative humidity, rainfall, air pressure, and seismicity. However, the contribution of each factor, and how the dominant factor affects seasonal $\delta v/v$, remain unclear between different types of slopes. We herein conduct 2 years of in situ monitoring of daily $\delta v/v$ on the Pubugou (PBG) and Jinpingzi (JPZ) slopes. We then employ a tree-based ensemble tool (XGBoost) to predict $\delta v/v$ from these factors, and apply SHapley Additive exPlanation (SHAP) to quantify the contribution of each factor. The results suggest that the XGBoost model achieves R^2 values of 0.850 and 0.677 for the PBG and JPZ slopes, respectively. SHAP analysis highlights temperature as the dominant factor while revealing a contrasting pattern: a linear correlation on the PBG slope, but a nonlinear relationship on the JPZ slope. Additionally, compared to traditional multiple linear regression, the SHAP algorithm provides a more comprehensive quantification of nonlinear correlation. Our work demonstrates that combining machine learning with interpretability allows us to quantify the global and local impacts of each factor, supporting broader application of interpretable machine learning in fields from engineering to geophysics.

1. Introduction

Landslides are one of the most important evolutionary processes of material migration on the shallow surface in mountainous regions of southwest China (Hung et al., 2014), which often exhibit sudden and unpredictable acceleration phases and flows (Burbank et al., 1996; Clarke & Burbank, 2010), constituting major damage to human lives and property. In addition to the instant hazards caused by landslide failures, the potential geo-risks that slow-moving slopes pose to infrastructure and human safety are of greater concern (Lacroix et al., 2014; Shi et al., 2021; Ye et al., 2024). Moreover, extreme weather, possibly induced by global climate change, along with regional seismic activity, has the potential to exacerbate these risks (Stevenson et al., 2022; Stoffel et al., 2024), pushing them to more dangerous levels. This further complicates the relationship between the damage evolution of slopes and external forcings, such as air temperature, rainfall, and seismicity. To address this challenge, it is necessary to better understand the damage evolution patterns and mechanisms of slow-moving slopes under long-term external factors.

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Currently, the analysis of continuous seismic waveforms is one of the few economical and efficient detection methods that can penetrate subsurface media and obtain the physical state and evolution of the medium by monitoring seismic velocity changes ($\delta v/v$). Related experimental studies have demonstrated that the development of damage or the reduction of rigidity is related to the reduction of $\delta v/v$ (Bièvre et al., 2018; Larose et al., 2015; Mainsant et al., 2012; Niu et al., 2008; Xie et al., 2023). Therefore, $\delta v/v$ inferred from ambient noise interferometry, which uses the multiple scattering of weak vibrations by heterogeneities in the crust (Sens-Schönfelder & Wegler, 2006; Snieder, 2006), allows remote assessment of the damage state of slopes at depth.

Slow-moving slopes undergo dynamic evolution driven by various external factors, including air temperature, rainfall, relative humidity, air pressure, and seismicity (Comegna et al., 2013; Liu et al., 2024; Xie et al., 2023). These factors alter the stress state within the rock and soil mass, affecting the damage evolution process of slopes. As a high-sensitivity proxy characterizing the damage to subsurface media, $\delta v/v$ records the long-term evolutionary behavior of these factors over time. Although traditional multiple linear regression can assess their relationship by calculating response coefficients between different factors and $\delta v/v$ (Q.-Y. Wang et al., 2017), a quantitative investigation is lacking in the following two aspects: (a) the marginal contribution of different factors to $\delta v/v$ and (b) how the dominant factor affects $\delta v/v$. Moreover, comparative studies of different types of slopes (e.g., rock slope vs. clastic soil slope) under long-term external factors remain scarce.

Recently, machine learning has increasingly been applied to predict slope stability due to its ability to handle complex and nonlinear data relationships (Chang et al., 2023; Liu et al., 2021; Schaller et al., 2025; Weidner et al., 2019). Existing studies mainly employ classical supervised machine learning algorithms, such as Random Forest (RF), eXtreme Gradient Boosting (XGBoost), and Light Gradient Boosting Machine (LightGBM), to construct landslide susceptibility models (Chang et al., 2023; Lin et al., 2022; Sun et al., 2024; Zhu et al., 2025). These models typically take various factors, such as elevation, lithology, soil type, and rainfall, as input features and use historical landslide data sets for training and testing, ultimately classifying stable and unstable slopes. However, such approaches only assess spatial susceptibility and fail to capture the time-dependent behavior of slope failures. In contrast, $\delta v/v$ provides long-term, high-temporal-resolution, and real-time monitoring of slope damage states, yet its integration with machine learning remains unexplored. Additionally, despite their impressive predictive power, most machine learning models remain difficult to explain the reasoning behind their predictions (Jiang et al., 2024). Specifically, while these models provide global feature importance analysis, they still lack the ability to interpret the predictions of local samples. Furthermore, inconsistencies in how different models define and quantify feature importance limit the comparability and universality of the resulting interpretations (S. S.-C. Wang et al., 2021).

Therefore, this study develops an interpretable machine learning (IML) model based on seasonal $\delta v/v$, focusing on two primary objectives: (a) quantifying the global and local contributions of long-term external factors to seasonal $\delta v/v$ and (b) comparatively studying the relationships between these factors and seasonal $\delta v/v$ across different types of slopes. The study centers on two representative slopes: the rock-based PBG slope and the clastic soil-based JPZ slope. For each area, we establish the XGBoost model, which is used to predict seasonal $\delta v/v$ by incorporating multiple physical variables that affect slope stability, including air temperature, relative humidity, air pressure, rainfall, and seismicity. To unveil the black-box nature of the model, we employ the SHapley Additive exPlanations (SHAP) algorithm, a post-hoc interpretability method (Lundberg & Lee, 2017). The method decouples complex nonlinear relationships between multiple physical variables and seasonal $\delta v/v$, thereby improving our understanding of the driving factors influencing $\delta v/v$.

2. Study Area and Data

2.1. Site Description

Figure 1a illustrates complex regional tectonic settings around the PBG and JPZ slopes in southwest China. The PBG slope is located within the deep alpine valleys of the Dadu River between the western margin of the Sichuan Basin and the eastern edge of the Tibetan Plateau (Del Gaudio et al., 2019). It is surrounded by large-scale active faults (black lines), including the Xianshuihe fault (XSHF), Longmenshan fault (LMSF), and Xiaojinhe fault (XJHF). From 1 October 2020, to 26 February 2023, a total of 18,339 earthquakes ($M_L \geq 1$) were recorded within a 200-km radius of the PBG slope. The JPZ slope, located on the eastern edge of the Tibetan Plateau near the Jinsha River valley, is also surrounded by active faults such as the Puduhe fault (PDHF) and Zemuhe-Xiaojiang fault (ZMH-XJF). Due to the relatively low fault activity around the JPZ slope, only 1,080 earthquakes ($M_L \geq 1$)

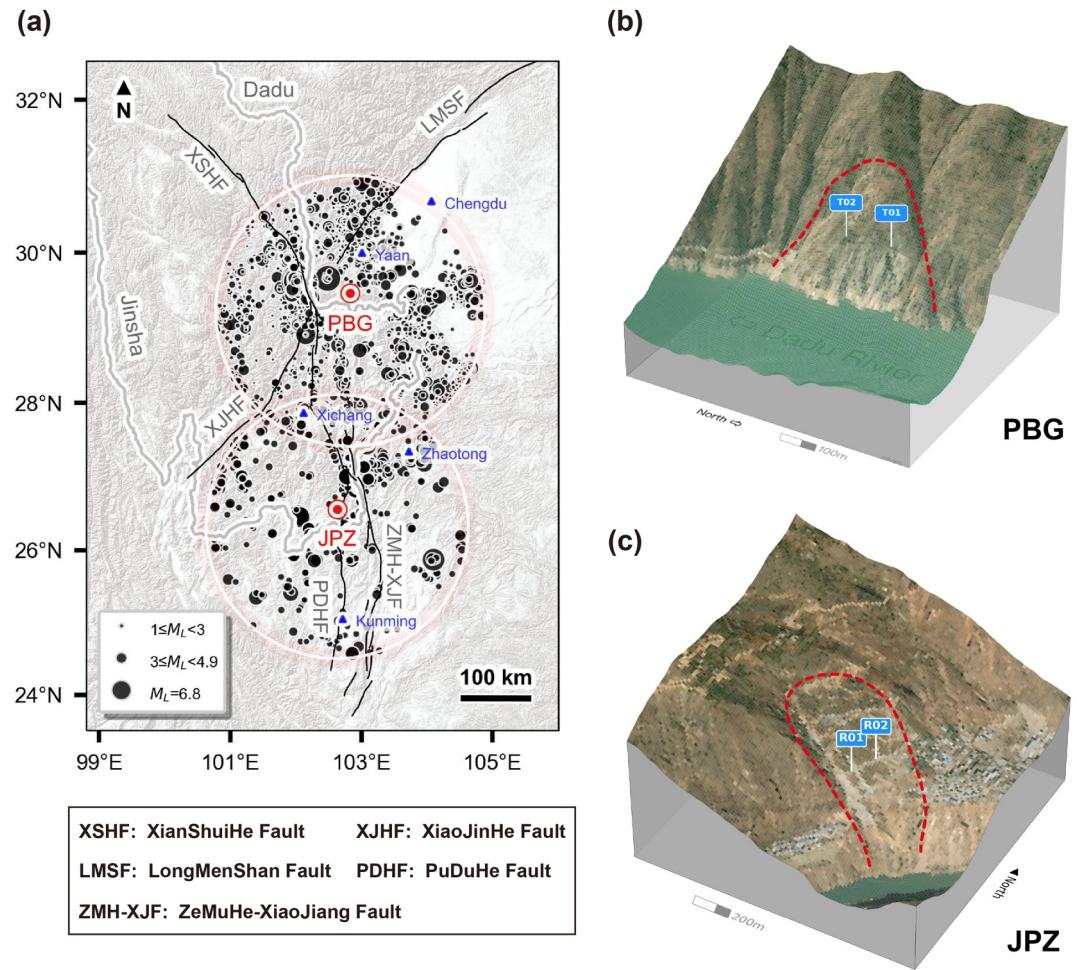


Figure 1. (a) Tectonic setting surrounding the Pubugou (PBG) rock slope and the Jinpingzi (JPZ) clastic soil slope. A total of 18,339 earthquakes ($M_L \geq 1$) were recorded within a 200-km radius around the PBG slope between 1 October 2020, and 26 February 2023. There are 1,080 earthquakes ($M_L \geq 1$) recorded within a 200-km radius around the JPZ slope between 13 July 2022, and 22 June 2024. (b) Three-dimensional terrain map of the PBG slope generated from the digital elevation model (DEM). The red dashed line delineates the boundary of the PBG slope, and blue fields indicate the locations of two seismic stations (T01 and T02). (c) Three-dimensional terrain map of the JPZ slope generated from the DEM. The red dashed line delineates the boundary of the JPZ slope, and blue fields indicate the locations of the two seismic stations (R01 to R02).

were recorded between 13 July 2022, and 22 June 2024. Note that the earthquake catalog is sourced from the Sichuan and Yunnan Earthquake Administration networks.

Figures 1b and 1c show the terrain maps of the PBG and JPZ slopes, located along the western banks of the Dadu River and Jinsha River, respectively. The PBG slope lies approximately 800 m from the dam of the PBG Hydropower Station (Xie et al., 2023), and the JPZ slope is about 2,500 m from the dam of the Wudongde Hydropower Station (Shu et al., 2017). Since both slopes are situated adjacent to the hydropower facilities, their stability is critical for the safe operation of the stations and for protecting downstream populations and infrastructure.

In addition, the PBG and JPZ slopes differ significantly in their topographical and geological characteristics, as described below. The PBG slope, with an elevation ranging from 780 to 1,180 m above mean sea level (a.m.s.l.) and an average slope angle of 45° , is composed of relatively complete tuffaceous rocks and diabases. The degree of weathering decreases with increasing depth, and intact rocks exist at an average depth of 70 m (Huang et al., 2021; Xie et al., 2023). Surface GNSS monitoring indicates progressive toppling of the upper part (>980 m) at an annual displacement rate of 21 mm/yr, while the displacement rate of the lower damaged part (<980 m) is approximately 3 mm/yr due to reinforcement with prestressed anchor cables (Huang et al., 2021; Yang

et al., 2013). We thus consider the unstable PBG slope as a slow-moving rock slope (Hungr et al., 2014). In contrast, the elevation of the JPZ slope ranges from 790 to 1,550 m (a.m.s.l.), with an average slope angle of 23°. The sliding mass of the slope consists of an upper layer of loose dolomite blocks and rubble, ranging from 20 to 150 m in thickness, and a lower layer of silty clay containing phyllite debris, ranging from 16 to 45 m in thickness (Shu et al., 2017; Zhao et al., 2018). The daily displacement rate of the slope varies between 69 and 318 mm/yr, showing a continuous subsidence trend (Shu et al., 2017). We thus classify the JPZ slope as a translational-type clastic soil slope (Glastonbury & Fell, 2008; Hungr et al., 2014). More detailed descriptions of the geological conditions of the PBG and JPZ slopes can be found in Shu et al. (2017), Zhao et al. (2018), Xie et al. (2023).

2.2. Seismic Data and Seismic Velocity Change Estimation

The seismic noise data consist of continuous recordings from two stations on the PBG slope, covering 879 days from 1 October 2020, to 26 February 2023, and from two stations on the JPZ slope, covering 711 days from 13 July 2022, to 22 June 2024. The two seismic stations (T01 and T02) on the PBG slope are deployed on flat and stable rocky spots in the central area (~990 m a.s.m.l.) of the slope, with an inter-station distance of ~157 m (Figure 1b). The two seismic stations (R01 to R02) on the JPZ slope are similarly deployed in the central area of the slope (~1,172 m a.s.m.l.), with an inter-station distance of ~169 m (Figure 1c). The base of each station is firmly coupled to the slope using a reinforced concrete basement, and the stations are further sheltered in small concrete huts. In such a way, the observation condition of the ambient seismic noise is improved, and the negative effects of the environment are reduced (i.e., wind, rain) and of the slope itself (i.e., unstable area). Furthermore, each station is equipped with an identical three-component short-period seismometer providing a relatively flat frequency response between 0.5 and 20 Hz. The seismometers are digitized by a 24-bit data acquisition system at a 100 Hz sampling rate, with a system clock error of less than 10 μ s. Continuous seismic records are then stored locally in 1-hr long sequences.

Based on the raw seismic noise spectrograms (see Figure S1 in Supporting Information S1), hourly continuous seismic records are band-pass filtered between 2 and 20 Hz. We then apply spectral whitening and one-bit normalization to homogenize the frequency band, and finally clip transient interference signals with amplitudes exceeding three standard deviations in the time domain. To reconstruct the wave propagation between the stations, we use the pre-processed seismic records to compute hourly Cross-Correlation Functions (CCFs) for the vertical components in each station pair, with a time lag of 15 s. We then average the causal and acausal parts of each CCF and stack the 24-hr CCFs to obtain the daily CCFs (Figures S2a and S2b in Supporting Information S1). This improves the stability of the daily CCFs and enhances the signal-to-noise ratio (Bièvre et al., 2018; Q.-Y. Wang et al., 2017). By stacking all daily CCFs for a given station pair, a reference CCF can be obtained (Figures S2c, i and S2d, i in Supporting Information S1). We select a lag time of 1 s as the beginning of the coda window to exclude the influence of the direct surface wave arrivals. The energy envelope of the reference CCF (Figures S2c, ii and S2d, ii in Supporting Information S1) shows that the energy decays to 10 s and then fluctuates steadily, marking the end of the coda window (Tchawé et al., 2020). Assuming the medium undergoes a homogeneous change, the travel-time perturbation in the coda is related to the relative change in the seismic velocity by $\delta v/v = dt/t$ (Snieder, 2006), indicating stretching or compression of the waveform following a decrease or increase in velocity. We therefore apply the stretching method to measure the $\delta v/v$ in the coda window (Sens-Schönfelder & Wegler, 2006; Xie et al., 2023), as shown in Figure 2i. Note that the depth sensitivity of the reconstructed CCFs in the 2–20 Hz frequency range, which characterizes $\delta v/v$ from a few meters to approximately 200 m below the ground surface (Xie et al., 2023), is well-suited for our study on damage to shallow slopes. Furthermore, periodical amplitude changes of the ambient seismic noise in the 2–20 Hz frequency band are not observed across the stations at seasonal scales (Figure S3 in Supporting Information S1).

2.3. Multi-Physical Variable Measurements

To study the impact of long-term fluctuations in various physical variables on slope stability, we deployed meteorological and rainfall sensors at the central part of the two slopes (T01 and T02, R01, and R02), as shown in Figures 1b and 1c. These sensors continuously record air temperature, rainfall, relative humidity, and air pressure at an hourly sampling rate. To match the daily resolution of the $\delta v/v$ measurements for comparison, the data were resampled daily and averaged across the two stations. Moreover, we collect observations of peak ground velocity (PGV) and daily number of earthquake events (earthquake frequency) during periods of local seismic activity, using the earthquake catalog and seismic data, where the latter are recorded by the stations described in

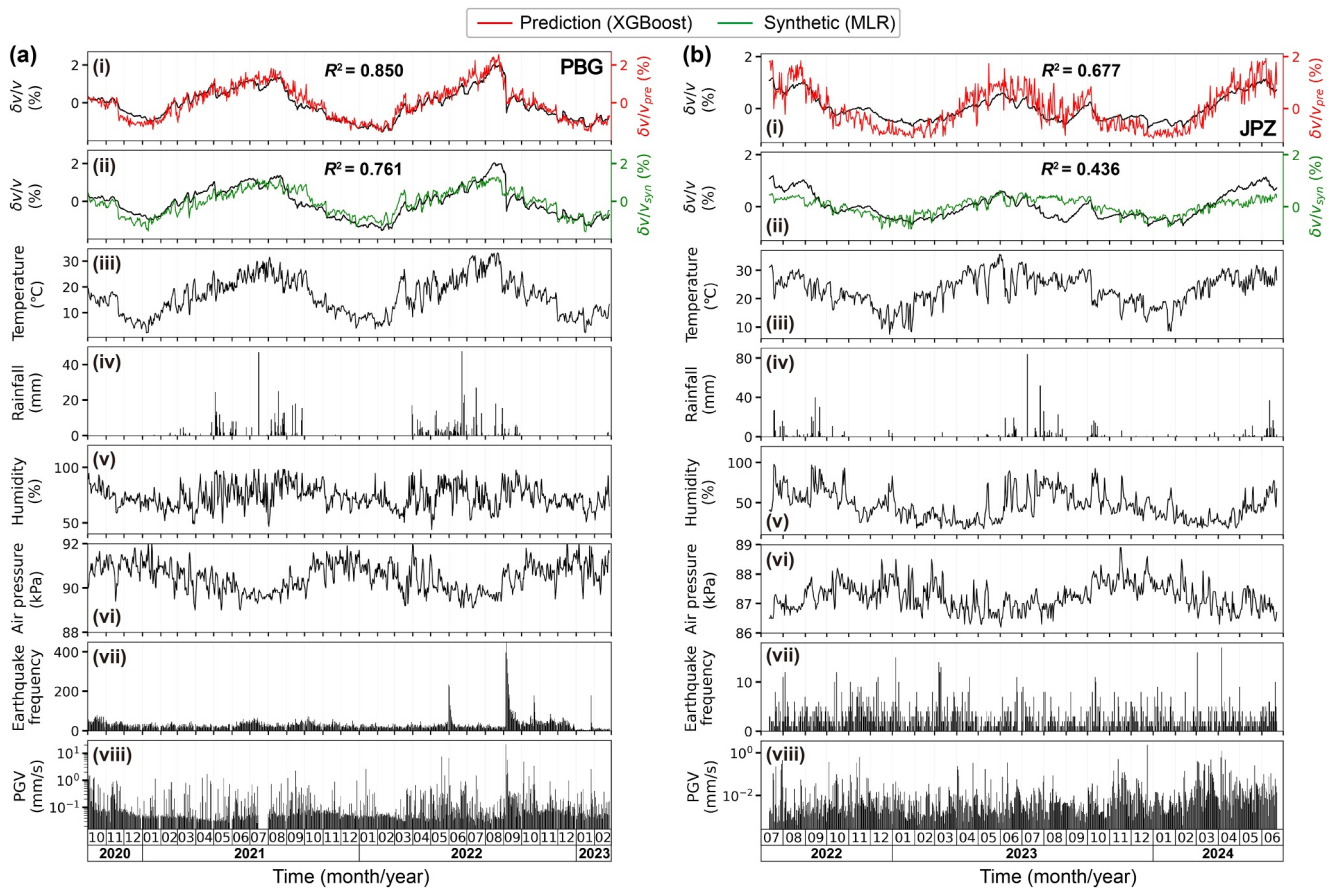


Figure 2. (a) Long-term $\delta v/v$ and multi-physical variables recorded at a 1-day sampling rate on the Pubugou slope between 1 October 2020, and 26 February 2023 (879 days). (b) Long-term $\delta v/v$ and multi-physical variables recorded at a 1-day sampling rate on the Jinpingzi slope between 13 July 2022, and 22 June 2024 (711 days). For both slopes, the variables are: (i) the measured $\delta v/v$ (black) and the predicted $\delta v/v$ (red) obtained from the XGBoost model; (ii) the measured $\delta v/v$ (black) and the synthetic $\delta v/v$ (green) fitted using a multiple linear regression method; (iii) air temperature; (iv) rainfall; (v) relative humidity; (vi) air pressure; (vii) earthquake frequency (daily number of earthquake events); (viii) peak ground velocity.

Section 2.2. These are potential drivers of subsurface seismic velocity changes (Poli et al., 2020; Q.-Y. Wang et al., 2021), as recurring seismic shaking can progressively weaken the soil by fracturing the rock mass and thereby affect seismic velocities at depth (Bontemps et al., 2020; Keefer, 1984; Meunier et al., 2008). Finally, we integrate them into the multi-physical variable data set (Figure 2), facilitating the analysis of their correlation with $\delta v/v$ over time. Details on the calculation of PGV can be found in Text S1 in Supporting Information S1.

3. Methods

3.1. XGBoost Model Prediction

Multi-physical variables interacting within slopes exhibit complex nonlinear relationships that can be explored using machine learning models (Choubin et al., 2019; Hasan et al., 2023; Liu et al., 2021; Yin et al., 2021). In this study, we use XGBoost, an ensemble learning model renowned for its ability to model complex nonlinear functions, to incorporate the multi-physical variables (commonly referred to as “input features”) for predicting $\delta v/v$ (Chen & Guestrin, 2016). The model is capable of minimizing the risk of overfitting and maintains strong generalization performance, due to its depth-wise tree growth strategy and robust regularization mechanism (Amerthen & Praba, 2025; Chen & Guestrin, 2016).

As the architecture and performance of the model are controlled by hyperparameters, which are listed in Table S1 in Supporting Information S1, we perform these hyperparameters tuning using a random search approach coupled

with five-fold cross-validation on 1,000 combinations of parameters for the training set (totaling 5,000 fits) (Hasan et al., 2023). This tuning aims to find the optimal set of hyperparameters for each slope (Table S1 in Supporting Information S1) to minimize the validation error and thereby enhance the model's generalizability (Lu et al., 2023). Moreover, we normalize the data set by subtracting the mean and dividing by the standard deviation to eliminate the effects of differences in scale among input features. The normalized data set is then randomly divided into training and testing sets, with a 75%–25% split, from which we analyze the statistical metrics, including the coefficient of determination (R^2), root-mean-squared error (RMSE), and mean absolute error (MAE) (Wei et al., 2020). To evaluate the performance of XGBoost model, we compared it with several widely-used machine learning algorithms, including RF and LightGBM. It should be noted that these models are used to capture the relationships between the input features and $\delta v/v$, rather than to model temporal prediction.

3.2. SHAP Explainability Analysis

The core objective of this study is not only to predict $\delta v/v$, but to identify the most influential environmental factors and to elucidate their relationships with the observed $\delta v/v$ variations by leveraging model interpretability and visualization tools. To achieve this, we employ the SHAP algorithm to investigate the decision-making mechanisms underlying the XGBoost model (Lundberg & Lee, 2017; Lundberg, Nair, et al., 2018; Ma et al., 2023; Reddy et al., 2023). It mathematically quantifies each input feature's contribution to the predicted $\delta v/v$ by the XGBoost model through the construction of an “explanation model.” The explanation model represents the XGBoost predictions as the sum of the mean predicted value and the contribution of each input feature, expressed mathematically as:

$$y_i = y_{base} + f(x_{i,1}) + f(x_{i,2}) + \dots + f(x_{i,j}), \quad (1)$$

where $f(x_{i,j})$ is the contribution of the i th sample in the j th feature to y_i (also known as the “SHAP value”), y_{base} is the baseline (the mean prediction of the model when no input features are provided), and y_i is a predicted value of the model for this sample.

Other commonly used interpretability methods, such as the gain method or split count (Lundberg, Erion, & Lee, 2018; S. S.-C. Wang et al., 2021), provide only global variable importance, but fail to reveal the complex and localized interactions between each sample of each feature and $\delta v/v$. In contrast, SHAP assigns a contribution value to each sample for each feature, allowing the evaluation of local variable importance while yielding relatively stable global variable importance metrics. Specifically, the average of the absolute SHAP values for each feature is interpreted as a measure of global variable importance, expressed as $g(x_j) = (\sum_{i=1}^n |f(x_{ij})|)/n$ (Molnar, 2020). Although the tree-based SHAP interpreter is emerging as a de facto method for explaining the outputs of machine learning models (Feng et al., 2024; Merrick & Taly, 2020), interpretation still requires caution and domain expertise (I. E. Kumar et al., 2020).

4. Results and Discussions

4.1. Model Performance Evaluation

We compare the XGBoost model with RF and LightGBM in predicting $\delta v/v$, as described in Section 3.1. We then compute the statistical metrics of each model on both training and testing sets for the PBZ and JPZ slopes. These metrics, including R^2 , RMSE, and MAE, are detailed in Tables S2 and S3 in Supporting Information S1. The results show that the XGBoost model achieves R^2 values of 0.850 and 0.677 on the testing set for the PBZ and JPZ slopes, respectively, and exhibits higher R^2 and lower errors (RMSE and MAE) than RF and LightGBM. Moreover, the predictions from the XGBoost model on the full data set match well with the $\delta v/v$ measurements, as shown in Figures 2a,i and 2b,i. As described above, the XGBoost model has demonstrated robust performance in predicting $\delta v/v$ across different types of slopes.

4.2. Contribution of Input Features to the $\delta v/v$ on Different Slopes

To decouple nonlinear interactions between input features and long-term $\delta v/v$, we perform a comprehensive interpretability analysis of the XGBoost model using SHAP values (denoted as $\text{SHAP}_{\delta v/v}$), including global

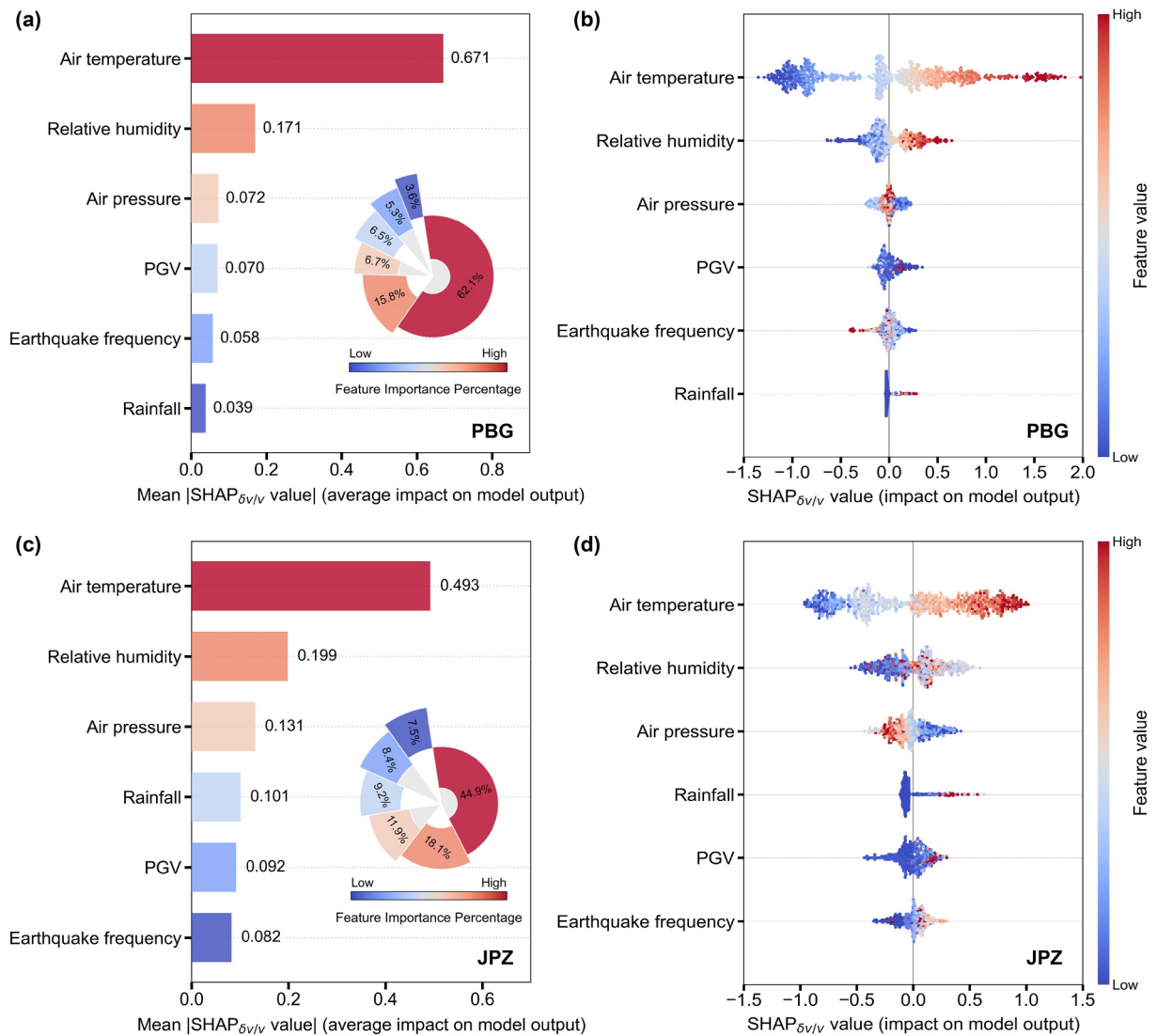


Figure 3. (a, c) Show the rankings of the mean absolute SHAP_{δv/v} values for the six input features on the Pubugou (PBG) and Jinpingzi (JPZ) slopes, respectively. Rose charts in (a) and (c) illustrate the contribution of each input feature to $\delta v/v$ using the percentage method. (b, d) Show the SHAP_{δv/v} value distributions for each input feature according to the feature importance ranking.

feature importance and local feature dependence. Note that a positive SHAP_{δv/v} value indicates that a feature increases the predicted $\delta v/v$, whereas a negative SHAP_{δv/v} value indicates that it decreases the predicted $\delta v/v$.

We first examine global feature importance in the XGBoost model to identify the most influential factors and assess the extent of their influence on the $\delta v/v$. By computing the mean absolute SHAP_{δv/v} values, we rank the global importance of six input features on the PBG and JPZ slopes (Figures 3a and 3c). In addition to feature ranking, we also show the impact trend of each feature on the $\delta v/v$ (Figures 3b and 3d). We observe that air temperature exhibits the highest feature importance and the widest distribution for both slopes, accounting for 62.1% and 44.9% of contributions to the PBG and JPZ slopes, respectively. This indicates its dominant role in controlling the $\delta v/v$. The corresponding SHAP_{δv/v} values increase with rising temperature, suggesting that higher temperature leads to larger $\delta v/v$.

Relative humidity consistently emerges as the second most important feature across two types of slopes. Interestingly, it shows a positive correlation with the $\delta v/v$ on the PBG slope (Figure 3b), whereas the relationship appears more complex on the JPZ slope (Figure 3d). Other features exhibit different importance rankings on the two slopes, though SHAP_{δv/v} value distributions for some features align with the existing understanding. For

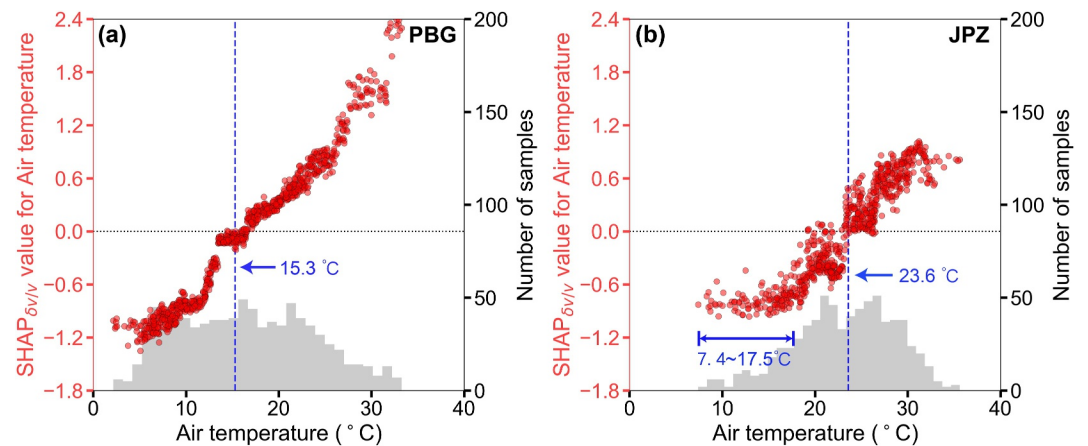


Figure 4. (a, b) Indicate $SHAP_{\delta v/v}$ value contributions of air temperature to $\delta v/v$ on the Pubugou and Jinpingzi slopes, respectively. The gray dashed horizontal line indicates the transition from negative to positive $SHAP_{\delta v/v}$ values. The gray histogram in each subplot shows the distribution of feature values in the entire data set.

example, rainfall has a greater influence on the clastic soil slope (JPZ) than on the rock slope (PBG). Given the limited interpretability of $SHAP_{\delta v/v}$ values for low-contribution features, we focus on the most important input feature (i.e., air temperature) and its influence on the $\delta v/v$.

We further investigate local feature dependence by comparing $SHAP_{\delta v/v}$ values with the original feature values. As shown in Figure 4, we observe a strong positive correlation between temperature and its corresponding $SHAP_{\delta v/v}$ values on both slopes. However, the nature of this correlation differs: it is linear on the PBG slope but nonlinear between approximately 7.4°C and 17.5°C on the JPZ slope. The inflection points, where $SHAP_{\delta v/v}$ values transition from negative to positive, also differ, occurring at approximately 15.3°C for the PBG slope and 23.6°C for the JPZ slope.

4.3. Interpretation of the $\delta v/v$ to the Air Temperature

Our study visually demonstrates that daily $\delta v/v$ time series dominated by seasonal and long-term trends are mainly driven by the temperature field around the slopes, showing a strong positive correlation, consistent with previous studies (Bièvre et al., 2018; Liu et al., 2024; Xie et al., 2023). This correlation is largely attributed to thermally induced confinement stress within the medium caused by temperature forcing (Fokker et al., 2024; Guillemot et al., 2022). Such thermoelastic stress may increase the effective elastic moduli of the rock by reducing the contact between grains, leading to an increase in $\delta v/v$ with rising temperature. Nevertheless, an increase in $\delta v/v$ due to air temperature could also imply a development of the damage state. Thermal expansion-induced tensile stress can trigger exfoliation of outer rock layers, potentially causing rockfalls in steep terrains (Collins & Stock, 2016; Collins et al., 2018; Colombero et al., 2021). Although temperature changes themselves typically penetrate only tens to hundreds of centimeters below the ground surface (Bottelin et al., 2013), thermoelastic strain can extend deeper (even up to 68 m) inside the slope body (Gischig et al., 2011; Liu et al., 2024; Watson et al., 2004).

In our analysis, the PBG slope, composed of stratified volcanic rock, has higher thermal conductivity and thermal expansion than the JPZ slope, which consists of clastic soil, due to its greater overall compactness (El Sayed, 2011; Schärli & Rybach, 1984). This implies that the thermoelastic strain induced by the same temperature change can yield a more pronounced effect on $\delta v/v$ within the PBG slope, resulting in a lower inflection point of the temperature (Figures 4a and 4b). The JPZ slope has more complex lithological characteristics, comprising various geomaterials such as gravelly soil, silty clay, and phyllite debris. Thus, a nonlinear relationship is observed between the temperature and $SHAP_{\delta v/v}$ values on the JPZ slope (Figure 4b). Unlike previous landslide cases (Bièvre et al., 2018), both slopes are located in a typical concurrent rainfall-heat climate (Figure 2). Given the lack of precipitation events during low temperature (7.4°C to 17.5°C , Figures 2iii and 2iv) and the negligible impact of groundwater level changes (Shu et al., 2017), this nonlinear behavior is likely unrelated to hydrological processes. Instead, it is more plausibly attributed to the heterogeneous pore-scale structures and differences in

thermophysical properties among the various geomaterials within the JPZ slope. Moreover, we validate the robustness of the SHAP algorithm for interpreting the XGBoost model by incorporating additional features derived from physical model-based analysis, as detailed in Text S3 in Supporting Information S1.

4.4. Implication of Multiple Linear Regression

Compared to advanced machine learning algorithms, multiple linear regression (MLR) remains a widely used classical statistical technique for prediction and interpretation in various fields, such as groundwater table depth, landslide velocity, and local sea level, as it effectively decouples the dependencies between a response variable and multiple explanatory variables (Buchanan & Triantafyllis, 2009; V. Kumar et al., 2018; Krkač et al., 2020). Despite being a linear model, MLR can still achieve acceptable performance in predicting nonlinear behavior, even compared to more complex models (Barrero et al., 2006; Krkač et al., 2020; Q.-Y. Wang et al., 2017). Therefore, we select the three input features with relatively high contributions (i.e., air temperature, relative humidity, and air pressure) to summarize their influences by considering a linear relationship between $\delta v/v$ and the combined effects of these features, as expressed in Equation 2.

$$\frac{\delta v}{v}_{syn} = \alpha_t \cdot T + \alpha_h \cdot H + \alpha_p \cdot P + C, \quad (2)$$

where α_t , α_h , and α_p are response coefficients for the time series of air temperature (T), relative humidity (H), and air pressure (P), respectively, and C is a constant representing offset parameters. The best-fit values for α_t , α_h , and α_p are detailed in Table S4 in Supporting Information S1 on the PBG and JPZ slopes. Compared to the XGBoost model, MLR provides a notably inferior fit to the seismic velocity changes, as shown in Figures 2a, ii and 2b, ii.

According to these coefficients, air temperature (α_t) is positively correlated with $\delta v/v$, and α_t value is larger on the PBG slope than on the JPZ slope, which are consistent with the distribution of SHAP $_{\delta v/v}$ values on both slopes (Figure 3). The coefficient (α_p) of air pressure also aligns well with the SHAP $_{\delta v/v}$ results in the two primary aspects. Specifically, (a) the absolute value of α_p on the PBG slope is smaller compared to that on the JPZ slope (Figures 3a and 3c) and (b) the negative α_p value observed on the JPZ slope indicates a negative correlation between air pressure and $\delta v/v$ (Figure S4b in Supporting Information S1).

However, for relative humidity, α_h value on the PBG slope is higher than on the JPZ slope, which is inconsistent with the results of the SHAP algorithm (Figures 3a and 3c). This discrepancy may be due to the nonlinear relationship between humidity and $\delta v/v$ on the JPZ slope (Figure S5b in Supporting Information S1), which causes the linear regression coefficients to underestimate its actual influence. In contrast, the SHAP algorithm provides a more comprehensive quantification of nonlinear correlation, revealing a higher contribution of humidity to the JPZ slope. Therefore, interpretable models (e.g., the SHAP algorithm) have the potential to better understand the relationships between slope failure kinematics and their driving factors through quantifiable nonlinear insights, thereby offering more reliable decision support for landslide monitoring and early warning.

5. Conclusions

We conduct a 2-year monitoring of seismic velocity changes ($\delta v/v$) using ambient noise interferometry on two representative slopes in southwest China: the rock-dominated PBG slope and the clastic soil-based JPZ slope. The XGBoost model is then applied to predict seasonal $\delta v/v$ using external factors as input features, including air temperature, relative humidity, rainfall, air pressure, PGV, and daily number of earthquakes. The model's predictions are interpreted using the SHAP algorithm to analyze the relationships between these factors and $\delta v/v$. The results indicate that the XGBoost model achieves higher R^2 and lower errors (RMSE and MAE) on both slopes compared to classical models, such as RF and LightGBM. SHAP analysis reveals that air temperature is the dominant factor influencing seasonal $\delta v/v$, showing a robust positive correlation. A key finding is the contrasting pattern of this correlation: a linear relationship on the PBG slope versus a nonlinear relationship on the JPZ slope. This discrepancy is more likely attributed to the inherent material properties across slope types. Our work contributes to a better understanding of how long-term external factors drive slope mechanics and provides a new perspective for statistical analysis of different influencing factors in geologically hazardous slopes.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The monitored data and the code used in this study are available via (Li et al., 2026).

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